CSL 7640: NATURAL LANGUAGE UNDERSTANDING

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Programming-Assignment2:-

Report for Q.1)

1. INTRODUCTION:

The objective of this assignment is to learn word embeddings using textual data collected from IIT Jodhpur sources. Word embeddings represent words in a continuous vector space such that semantically similar words are placed closer together.

In this assignment, Word2Vec models were implemented from scratch, including both Continuous Bag of Words (CBOW) and Skip-gram with Negative Sampling. These models were then analyzed using similarity, analogy, and visualization techniques.

2. DATASET PREPARATION:

Textual data was collected from multiple IIT Jodhpur sources:

- Academic regulations document (mandatory)
- CSE department webpage
- Faculty profiles (AI department)
- Course curriculum

Only English text was considered, and irrelevant content was removed.

2.1 Preprocessing Steps

The dataset was cleaned using the following steps:

- Removal of boilerplate and formatting text
- Conversion to lowercase
- Tokenization (splitting into words)
- Removal of punctuation and non-text elements
- Stopword removal (e.g., “the”, “is”, “and”)

2.2 Dataset Statistics

- Number of documents: 4
- Total tokens: 1562
- Vocabulary size: 499

These values indicate a reasonably diverse dataset suitable for training word embeddings.

Fig 1: Raw Dataset Text Files

corpus.txt

calable systems mathematics courses such linear algebra calculus probability statistics and differential equations form the analytical foundation for engineering and data science applications students also study
undamental science subjects including electromagnetism optics and chemistry these subjects help understanding the physical principles behind engineering systems advanced computer science topics include algorithm design
nd analysis computer organization compiler design and programming languages these courses focus both theoretical concepts and practical implementation the curriculum also includes systemsrelated courses such systems
hinking systems modeling systems design and dynamical systems these subjects help students understand complex systems and their behavior courses optimization and operations research teach students how solve realworld
roblems efficiently using mathematical techniques students gain knowledge interdisciplinary subjects such socioeconomic networks business dynamics and management these areas connect technology with realworld
pplications and decision making students are involved research activities and academic learning students perform research and participate academic projects students engage research work and learning tasks courses
nclude exams assignments and evaluations courses involve exams and academic assessments courses require exams and practical evaluations phd students conduct research and publish papers phd programs focus research and
nnovation phd candidates work research problems and academic studies learning connected research and academic growth learning helps students perform research and study effectively study and learning are closely relatec
esearch activities students take courses and attend classes courses help students learn and prepare for exams students complete courses and participate exams research involves study learning and experimentation researc
elated learning and academic development research requires study learning and practical work courses are related students and academic learning courses help students gain knowledge and skills courses prepare students
or exams and research students prepare for exams through courses and learning students learn through courses and participate exams students study through courses and research activities phd related research and advance
tudy phd programs require research and academic contribution phd students focus research and study learning and study are connected concepts education study leads learning and research development learning improves
hrough study and research handson learning important part the curriculum students participate projects workshops and laboratory work apply theoretical knowledge practical scenarios courses such computer programming
lectronics and engineering mechanics provide practical exposure through experiments and assignments projectbased learning emphasized throughout the program students work multiple projects including major projects later
emesters develop problemsolving and research skills courses are associated with exams and evaluations courses involve exams and academic assessments students take courses and appear for exams courses prepare students
or exams and evaluation research related learning and study study and learning are part research students perform research through study and learning phd related advanced research and study phd students conduct researc
nd academic study the curriculum also includes courses communication skills writing psychology and leadership these subjects help developing soft skills and professional abilities elective courses allow students explor
pecialized areas based their interests these may include advanced topics artificial intelligence data science and emerging technologies the overall curriculum designed balance theory practical learning and
nterdisciplinary knowledge prepares students for careers research industry and innovation students are encouraged apply their knowledge realworld problems through projects internships and collaborative work this
pproach helps developing both technical expertise and critical thinking skills
he academic programs iit jodhpur are designed develop skilled individuals who can understand and solve realworld problems across multiple disciplines the institute emphasizes both technical knowledge and personal
evelopment through academic and extracurricular activities the academic system follows semesterbased structure combined with credit system and relative grading students are expected engage continuous learning and
evelop analytical and problemsolving abilities the academic session divided into two main semesters and summer term each semester includes structured schedule consisting registration classes examinations and evaluatior
rocesses students are required register for courses per the academic calendar registration includes selecting courses based program requirements and late registration may involve penalties courses are structured based
redits which depend lecture tutorial and practical hours students earn credits after successfully completing courses with passing grades teaching conducted primarily english and courses may include lectures laboratory
ork projects and independent study components evaluation continuous and includes assignments quizzes and examinations the performance students assessed through grading system that uses letter grades such and these
rades reflect relative performance among students attendance plays important role academic evaluation students are expected attend classes regularly and meet minimum attendance requirements qualify for examinations
ostgraduate programs offered include msc mtech and phd programs these programs are designed provide advanced knowledge and research opportunities various disciplines admission postgraduate programs based academic
ualifications entrance examinations interviews depending the program students must meet minimum credit and grade requirements continue and successfully complete their programs academic performance measured using
ndicators such sgpa and cgpa the institute also provides opportunities for research projects and independent study enhance practical knowledge and innovation leave policies and academic flexibility are provided under
ertain conditions but students are expected maintain academic discipline throughout the program students participate research projects and academic activities research activities include machine learning data science
nd artificial intelligence phd students work advanced research problems and publish papers courses include evaluation through exams assignments and projects students prepare for exams and academic assessments regularly
he academic program supports research learning and innovation
he computer science and engineering department iit jodhpur plays important role advancing modern technology and addressing realworld challenges focuses research and innovation emerging areas computing and aims
ontribute technological development society the department committed strengthening collaboration between academia and industry encourages students work interdisciplinary projects and research problems that have
ractical relevance the vision the department achieve excellence research innovation and practical applications computer science aims become one the leading departments the country and gain international recognition
pecialized research areas the department promotes both fundamental and applied research computer science and related fields supports collaboration with national and international institutions through joint research
nternships and academic exchange students are encouraged explore realworld problems and develop innovative solutions through research and projectbased learning the department also focuses building strong startup
cosystem emerging areas computing modern teaching approaches such experiential learning online education and hybrid learning models are adopted improve the learning experience the goal prepare students solve future
echnological challenges ethical and responsible manner the department also emphasizes the role computing society and promotes awareness ethical and policyrelated aspects emerging technologies
aculty members iit jodhpur including researchers such dipanjan roy mayank vatsa richa singh and manish aggarwal are actively involved diverse research areas across computer science artificial intelligence and
nterdisciplinary domains their work contributes both academic research and realworld technological advancements research artificial intelligence major focus area the institute faculty members work machine learning deep
earning generative and explainable they also explore humancentered decision making and interpretability intelligent systems which are important for building trustworthy and ethical technologies natural language
rocessing another important domain where research includes dialogue systems multilingual models language understanding and applications healthcare and climaterelated problems these efforts aim improve communication
etween humans and machines several faculty members focus computer vision and image analysis their work includes facial recognition biometrics video understanding medical image analysis and multimodal learning these
echnologies are widely applied security systems healthcare diagnostics and automation healthcare and neurosciencerelated research also prominent areas include brain imaging using fmri and eeg computational neuroscience
rain connectivity and braininspired artificial intelligence researchers work understanding human cognition and developing intelligent systems inspired brain functions data science and analytics form another key researc
rea faculty members study largescale data processing pattern recognition and predictive modeling they also work uncertainty modeling and datadriven decision making for solving complex realworld problems there strong
esearch activity cybersecurity blockchain and privacypreserving systems topics include cryptography secure communication decentralized systems and blockchain technologies these areas are important for ensuring secure
nd reliable digital infrastructure distributed systems cloud computing and edge computing are also widely studied faculty members focus building scalable efficient and highperformance computing systems that support
odern applications such iot and smart infrastructure theoretical computer science and algorithms are another important domain research includes graph algorithms computational complexity optimization and game theory
hese foundational areas support advancements many applied fields computer science faculty members are also involved robotics and intelligent systems research includes bioinspired robotics braincomputer interfaces and
utonomous systems these technologies aim improve humanmachine interaction and automation various industries interdisciplinary research strongly encouraged iit jodhpur faculty members collaborate across domains such
conomics social sciences healthcare and engineering this approach helps solving complex realworld problems that require knowledge from multiple fields the institute promotes innovation and research through publications
atents and collaborations with industry and academic institutions faculty members actively mentor students research projects internships and thesis work helping them develop both technical and practical skills overall
he faculty iit jodhpur contribute vibrant research ecosystem that combines theoretical knowledge with practical applications preparing students address future technological challenges students participate research

Fig 2: Cleaned corpus

2.3 Word Cloud

A word cloud was generated to visualize frequently occurring words.

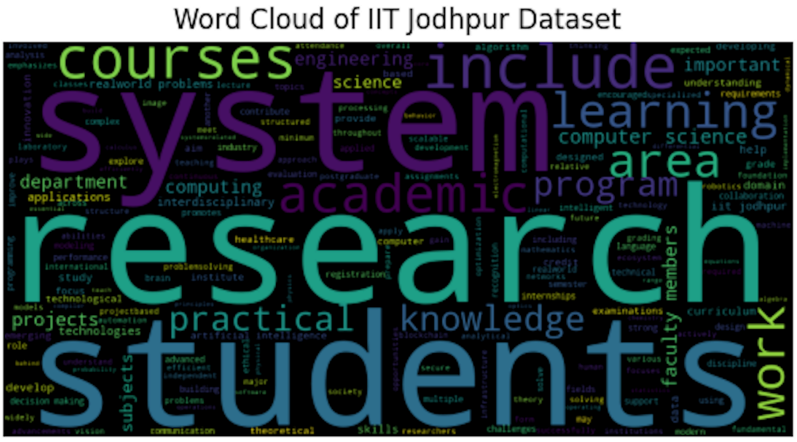


Fig 3: Word Cloud Visualization

3. MODEL TRAINING:

Two models were implemented from scratch:

- Skip-gram with Negative Sampling
- Continuous Bag of Words (CBOW)

3.1 Skip-gram Model

Skip-gram predicts context words given a target word. It is effective in capturing semantic relationships, especially in smaller datasets.

3.2 CBOW Model

CBOW predicts the target word from surrounding context words. It is faster but may capture fewer fine-grained relationships compared to Skip-gram.

3.3 Negative Sampling

Negative sampling was implemented to improve efficiency by updating only a subset of negative examples instead of the entire vocabulary.

3.4 Training Parameters

Parameter	Values Used
Embedding Dimension	50,100
Window Size	2, 4
Negative Samples	2, 5
Epochs	25

The loss was calculated as average loss per training sample, making it easier to interpret.

3.5 Training Results

The loss consistently decreased across epochs, indicating successful learning.

```
Training Skip-gram...
[Skip-gram] Epoch 1, Loss: 20.1358
[Skip-gram] Epoch 2, Loss: 15.1053
[Skip-gram] Epoch 3, Loss: 11.5635
[Skip-gram] Epoch 4, Loss: 9.2297
[Skip-gram] Epoch 5, Loss: 7.4828
[Skip-gram] Epoch 6, Loss: 6.0853
[Skip-gram] Epoch 7, Loss: 5.1258
[Skip-gram] Epoch 8, Loss: 4.3503
[Skip-gram] Epoch 9, Loss: 3.6834
[Skip-gram] Epoch 10, Loss: 3.2621
[Skip-gram] Epoch 11, Loss: 2.8728
[Skip-gram] Epoch 12, Loss: 2.5764
[Skip-gram] Epoch 13, Loss: 2.3047
[Skip-gram] Epoch 14, Loss: 2.1270
[Skip-gram] Epoch 15, Loss: 1.9245
[Skip-gram] Epoch 16, Loss: 1.8062
[Skip-gram] Epoch 17, Loss: 1.6856
[Skip-gram] Epoch 18, Loss: 1.5667
[Skip-gram] Epoch 19, Loss: 1.4826
[Skip-gram] Epoch 20, Loss: 1.4361
[Skip-gram] Epoch 21, Loss: 1.3832
[Skip-gram] Epoch 22, Loss: 1.3269
[Skip-gram] Epoch 23, Loss: 1.2693
[Skip-gram] Epoch 24, Loss: 1.2390
[Skip-gram] Epoch 25, Loss: 1.2216
```

Figure 4: Skip-gram Model Training Output

```
Training CBOW...
[CBOW] Epoch 1, Loss: 9.6608
[CBOW] Epoch 2, Loss: 8.9227
[CBOW] Epoch 3, Loss: 8.6711
[CBOW] Epoch 4, Loss: 8.3373
[CBOW] Epoch 5, Loss: 7.7524
[CBOW] Epoch 6, Loss: 7.2813
[CBOW] Epoch 7, Loss: 7.1914
[CBOW] Epoch 8, Loss: 6.8486
[CBOW] Epoch 9, Loss: 6.7670
[CBOW] Epoch 10, Loss: 6.5060
[CBOW] Epoch 11, Loss: 6.0723
[CBOW] Epoch 12, Loss: 5.7139
[CBOW] Epoch 13, Loss: 5.4781
[CBOW] Epoch 14, Loss: 5.2369
[CBOW] Epoch 15, Loss: 4.7367
[CBOW] Epoch 16, Loss: 4.5621
[CBOW] Epoch 17, Loss: 4.5728
[CBOW] Epoch 18, Loss: 4.1703
[CBOW] Epoch 19, Loss: 3.9408
[CBOW] Epoch 20, Loss: 3.6769
[CBOW] Epoch 21, Loss: 3.4982
[CBOW] Epoch 22, Loss: 3.4318
[CBOW] Epoch 23, Loss: 3.3393
[CBOW] Epoch 24, Loss: 3.1163
[CBOW] Epoch 25, Loss: 3.0421
```

Figure 5: CBOW Model Training Output

4. SEMANTIC ANALYSIS:

Semantic relationships were analyzed using cosine similarity.

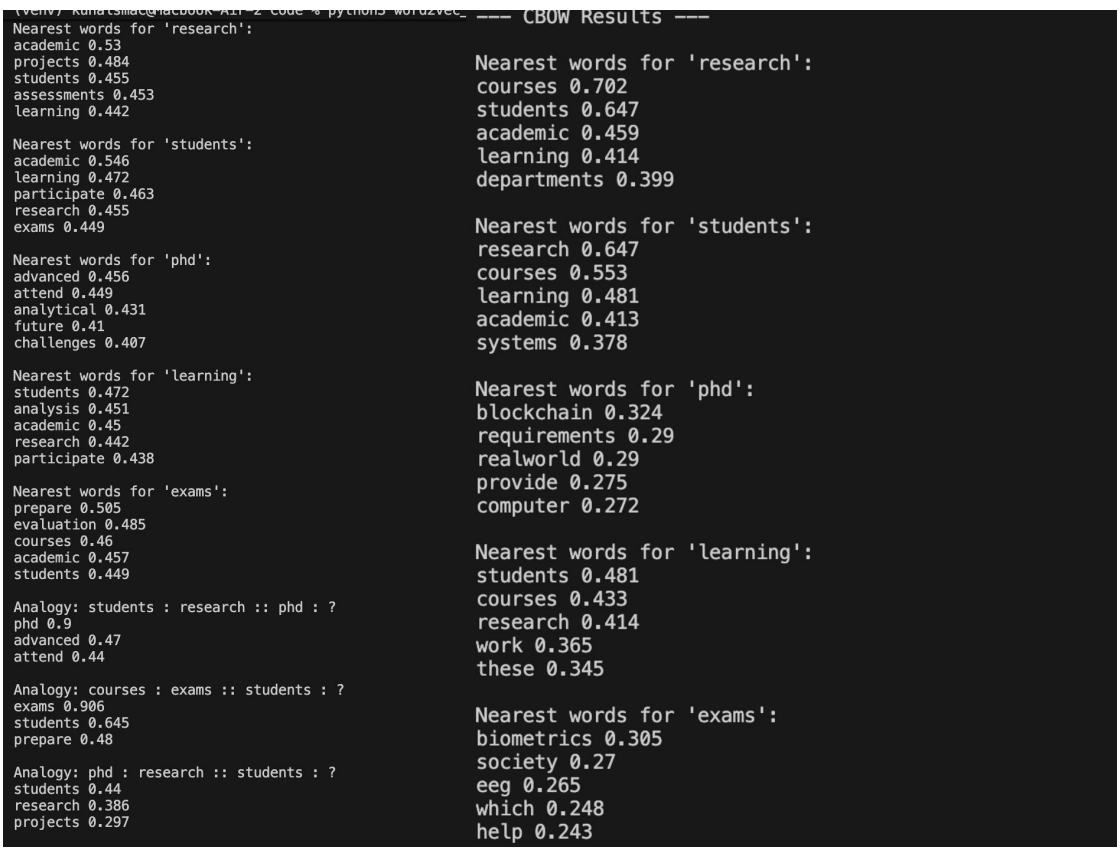
4.1 Nearest Neighbors

Top 5 nearest words were found for:

- research
- students
- phd
- learning

Observations:

- “research” is associated with academic and technical terms
- “students” relates to learning and projects
- “phd” relates to advanced study and research



```
Nearest words for 'research':
academic 0.53
projects 0.484
students 0.455
assessments 0.453
learning 0.442

Nearest words for 'students':
academic 0.546
learning 0.472
participate 0.463
research 0.455
exams 0.449

Nearest words for 'phd':
advanced 0.456
attend 0.449
analytical 0.431
future 0.41
challenges 0.407

Nearest words for 'learning':
students 0.472
analysis 0.451
academic 0.45
research 0.442
participate 0.438

Nearest words for 'exams':
prepare 0.505
evaluation 0.485
courses 0.46
academic 0.457
students 0.449

Analogy: students : research :: phd : ?
phd 0.9
advanced 0.47
attend 0.44

Analogy: courses : exams :: students : ?
exams 0.906
students 0.645
prepare 0.48

Analogy: phd : research :: students : ?
students 0.44
research 0.386
projects 0.297
```

```
--- CBOW Results ---

Nearest words for 'research':
courses 0.702
students 0.647
academic 0.459
learning 0.414
departments 0.399

Nearest words for 'students':
research 0.647
courses 0.553
learning 0.481
academic 0.413
systems 0.378

Nearest words for 'phd':
blockchain 0.324
requirements 0.29
realworld 0.29
provide 0.275
computer 0.272

Nearest words for 'learning':
students 0.481
courses 0.433
research 0.414
work 0.365
these 0.345

Nearest words for 'exams':
biometrics 0.305
society 0.27
eeg 0.265
which 0.248
help 0.243
```

Figure 6: Nearest Neighbor Words for Skip-gram and CBOW Models

4.2 Analogy Experiments

Three analogy experiments were conducted:

1. students : research :: phd : ?

2. courses : exams :: students : ?

3. phd : research :: students : ?

Observations:

- The model correctly identifies relationships between students, research, and academic activities
- Students and PhD both relate to research
- Courses and exams relate to students

```
Analogy: students : research :: phd : ?
phd 0.9
advanced 0.47
attend 0.44

Analogy: courses : exams :: students : ?
exams 0.906
students 0.645
prepare 0.48

Analogy: phd : research :: students : ?
students 0.44
research 0.386
projects 0.297
```

Figure 7: Analogy Results for Skip-gram Model

```
Analogy: students : research :: phd : ?
phd 0.9
advanced 0.47
attend 0.44

Analogy: courses : exams :: students : ?
exams 0.906
students 0.645
prepare 0.48

Analogy: phd : research :: students : ?
students 0.44
research 0.386
projects 0.297
```

Figure 8: Analogy Results for CBOW Model

4.3 CBOW vs Skip-gram Comparison

- Skip-gram produced stronger and more meaningful relationships
- CBOW produced relatively weaker but still valid results

This is expected because Skip-gram performs better on smaller datasets.

5. VISUALIZATION:

Word embeddings were visualized using PCA.

Two separate plots were generated:

- Skip-gram embeddings
- CBOW embeddings

5.1 Observations

- Words like “research”, “phd”, and “study” form clusters
- Academic-related words are grouped together
- Skip-gram shows more meaningful clustering than CBOW

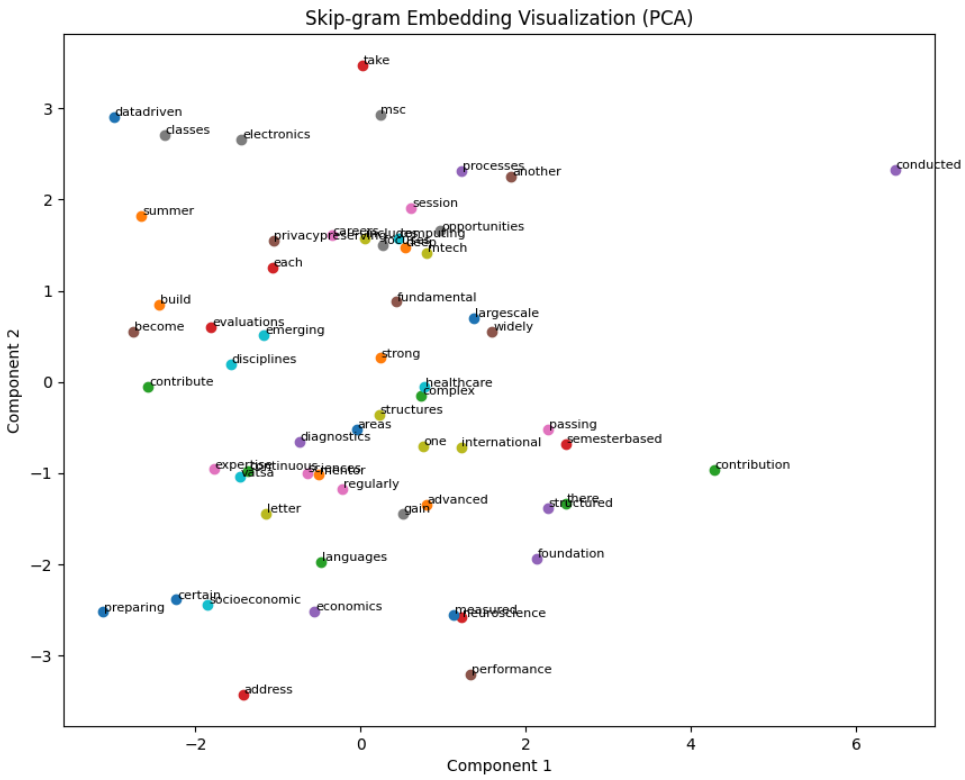


Figure 9: Skip-gram Embedding Visualization (PCA)

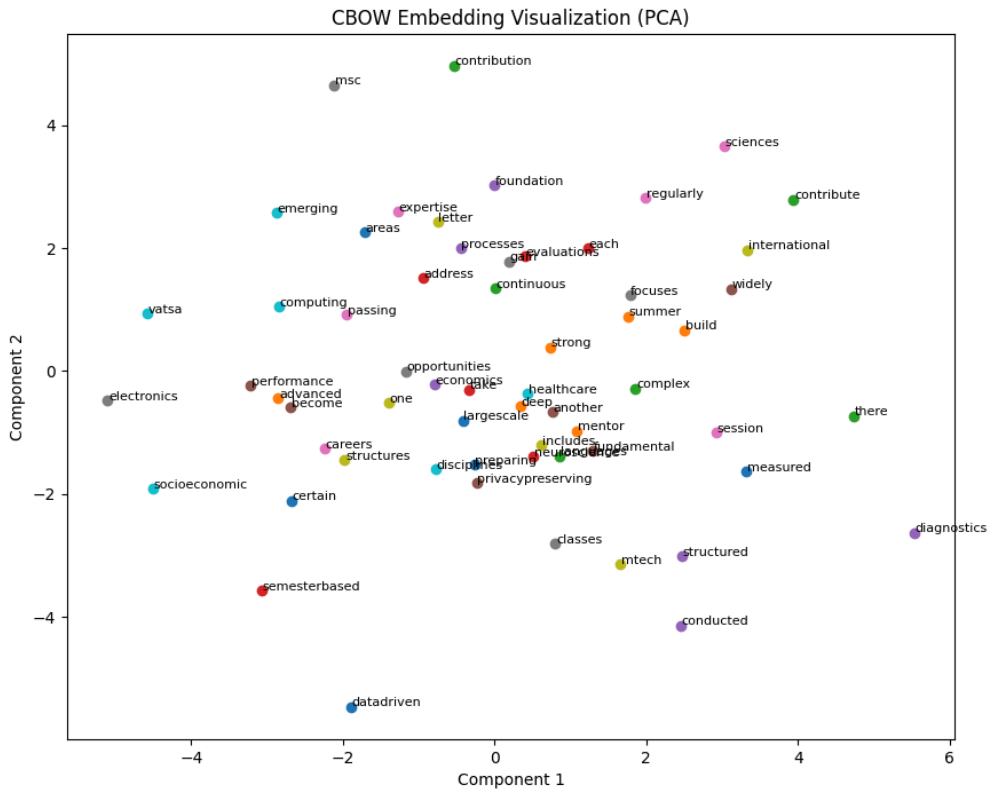


Figure 10: CBOW Embedding Visualization (PCA)

6. DISCUSSION:

- Increasing dataset size improved model performance
- Stopword removal reduced noise
- Skip-gram performed better than CBOW
- Negative sampling improved training efficiency

7. CONCLUSION:

In this assignment, Word2Vec models were implemented from scratch and trained on IIT Jodhpur data. The models successfully captured semantic relationships between words.

Skip-gram performed better than CBOW, and visualization confirmed meaningful clustering of related words.

This demonstrates that even simple implementations can effectively learn word representations when combined with proper preprocessing and training.